Rutuja Kadam

DAY07-IPv4 Addressing Part 2 & IP Configuration

THEORY (8 Marks)

Q1. What is a network address?

Answer: A network address identifies the entire network. Example: 192.168.10.0.

Q2. What is a broadcast address?

Answer: A broadcast address is used to send data to all hosts in a broadcast domain. Example: 192.168.10.255.

Q3. What is a usable host address?

Answer: A usable host address is an IP address that can be assigned to an individual device on a network.

Q4. What is the first usable IP address?

Answer: The first usable IP address is the address immediately after the network address. Example: 192.168.10.1.

Q5. What is the last usable IP address?

Answer: The last usable IP address is the address immediately before the broadcast address. Example: 192.168.10.254.

Q6. What does /24 mean?

Answer: /24 means that 24 bits represent the network portion of an IPv4 address.

Q7. What subnet mask corresponds to /24?

Answer: The subnet mask for /24 is 255.255.255.0.

Q8. What is a default gateway?

Answer: A default gateway allows a host to send data to destinations outside its local network. It is commonly a router.

Q9. Why does a router interface need an IP address?

Answer: A router interface needs an IP address to communicate with devices and forward traffic between different IP networks.

Q10. What is a static IP configuration?

Answer: Static IP configuration is a manually assigned, predictable IP configuration used to keep network devices reachable at known addresses.

Q11. What does "show ip interface brief" display?

Answer: It displays a quick summary of interfaces, including interface name, IP address, status and protocol.

Q12. What does "no shutdown" do conceptually on a Cisco interface?

Answer: The "no shutdown" command enables a Cisco interface administratively.

Q13. What does interface status "up/up" generally indicate?

Answer: Up/up indicates that the interface and its line protocol are operational.

Q14. Why should network configuration always be verified?

Answer: Network configuration should be verified to ensure that the settings are correct and connectivity is working as expected.

Q15. Why is IPv4 configuration important in cybersecurity?

Answer: IPv4 configuration is important in cybersecurity because security professionals use IP addresses, routers, firewalls, network logs, IDS/IPS alerts and SIEM events to monitor and protect networks. Incorrect configuration can cause connectivity problems and security monitoring errors.

PRACTICAL (4 Marks)

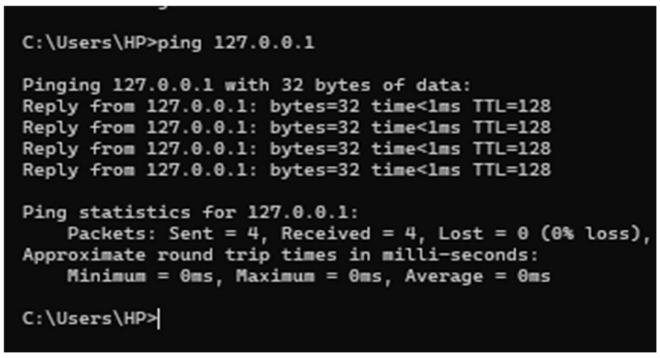
Record your own network configuration using ipconfig /all:

1. IPv4 Address: 192.168.0.101
2. Subnet Mask: 255.255.255.0
3. Default Gateway: 10.194.64.233
4. DNS Server: 192.168.0.1

SCREENSHOTS (2 Marks)

Attach screenshots of:

1)ping 127.0.0.1



2)ipconfig/all

```
C:\Users\HP>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-GQ12MPJH
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 16-AC-60-97-C6-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 16-AC-60-97-D6-65
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : MediaTek Wi-Fi 6E MT7922 (RZ616) 160MHz PCIe Adapter
   Physical Address. . . . . . . . . : 14-AC-60-97-E6-55
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2409:40c0:78:570a:42fd:53f:6726:305b(Preferred)
   Temporary IPv6 Address. . . . . . : 2409:40c0:78:570a:7dde:52c4:12fd:6072(Preferred)
   Link-local IPv6 Address . . . . . : fe80::49b1:753c:2f33:8f5e%8(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.194.64.242(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 13 November 2026 12:29:39
   Lease Expires . . . . . . . . . . : 13 November 2026 15:17:30
   Default Gateway . . . . . . . . . : fe80::dc53:cfff:fefb:f0b0%8
                                       10.194.64.233
   DHCP Server . . . . . . . . . . . : 10.194.64.233
   DHCPv6 IAID . . . . . . . . . . . : 85240928
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-46-78-B6-48-EA-62-93-48-CB
   DNS Servers . . . . . . . . . . . : 10.194.64.233
                                       2409:40c0:78:570a::fb
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 14-AC-60-97-E6-56
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 48-EA-62-93-48-CB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

3)ping <your default gateway> =ping 10.194.64.233

```
C:\Users\HP>ping 10.194.64.233

Pinging 10.194.64.233 with 32 bytes of data:
Reply from 10.194.64.233: bytes=32 time=6ms TTL=64
Reply from 10.194.64.233: bytes=32 time=17ms TTL=64
Reply from 10.194.64.233: bytes=32 time=14ms TTL=64
Reply from 10.194.64.233: bytes=32 time=17ms TTL=64

Ping statistics for 10.194.64.233:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 17ms, Average = 13ms

C:\Users\HP>
```

4)Packet Tracer topology (PC → Switch → Router), if completed.

TASK 4: PACKET TRACER LAB

PC Configuration:

- IPv4 Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Router Configuration:

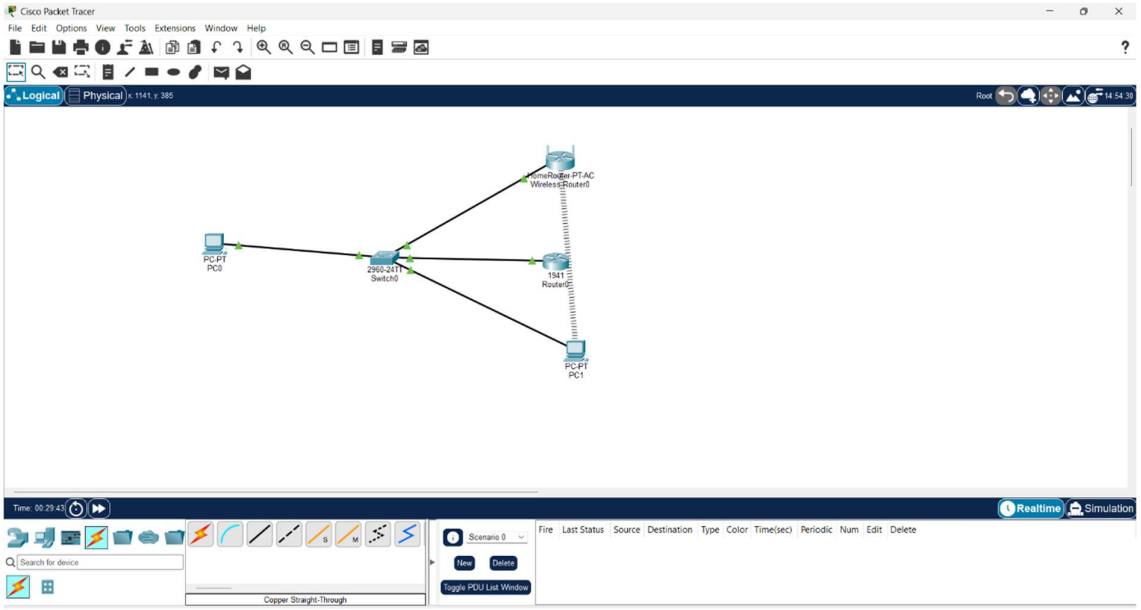
- enable
- configure terminal
- interface gigabitEthernet 0/0
- ip address 192.168.10.1 255.255.255.0
- no shutdown
- exit
- end
- show ip interface brief

Ping Test:

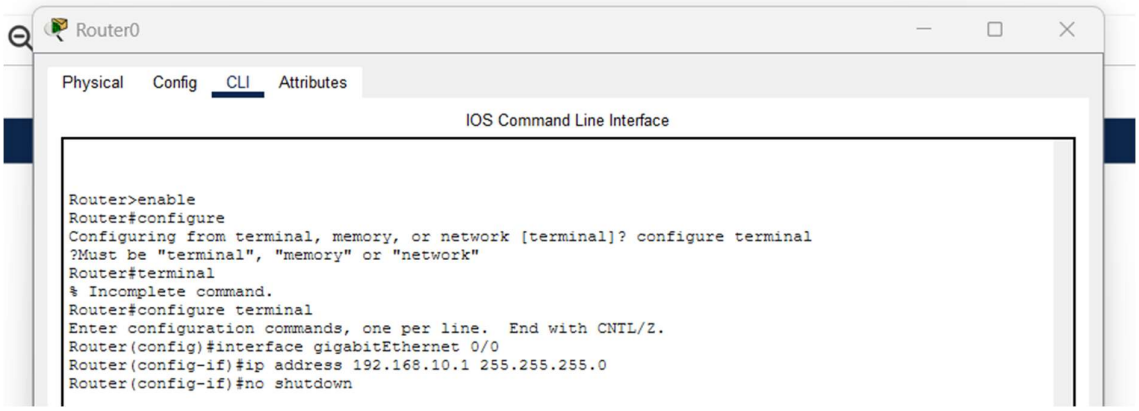
ping 192.168.10.1

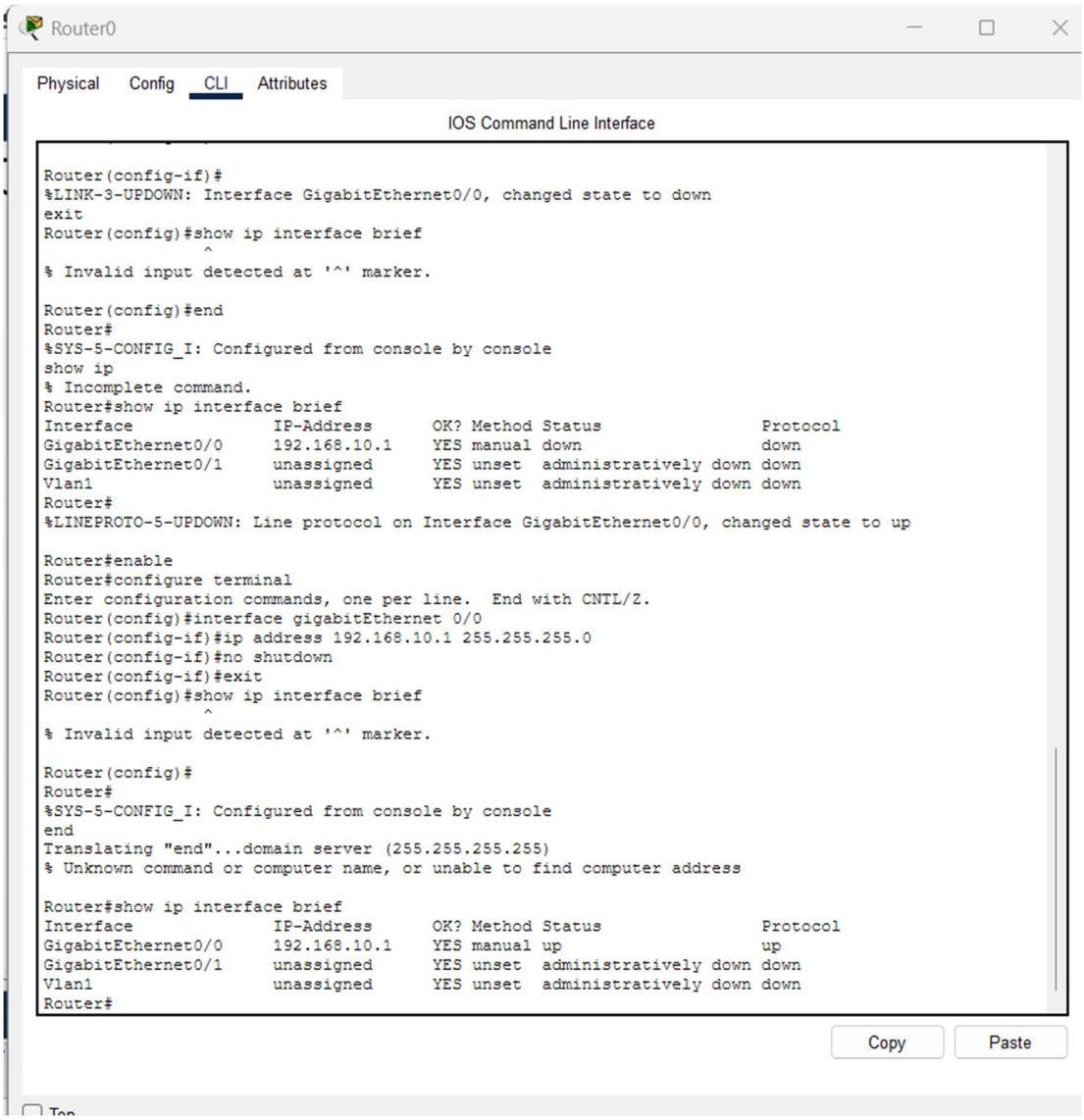
Expected Result: Successful ping with 0% packet loss.

Cisco Packet Tracer Network Topology (PC → Switch → Router)



Router Interface Configuration Verification using show ip interface brief





PC-to-Router Connectivity Test using ping 192.168.10.1

